# Advanced Quantitative Text Analysis

## Day 1: From Text to Measurement

Petro Tolochko, Noëlle Lebernegg

# Course Overview

1. **Day 1:** Text representation, preprocessing, feature engineering
2. **Day 2:** Dictionaries, supervised, and unsupervised classification
3. **Day 3:** Embeddings, transformers, LLMs
4. **Day 4:** Multilingual analysis, evaluation, research design, project talk

# Objectives

- Understand major approaches to automated text analysis
- Learn how text is transformed into data
- Identify practical challenges in real text-analysis projects
- Critically evaluate representation and modeling choices
- Connect methods to your own research questions

# Course Structure

- Theoretical input
- Practical sessions
- Discussion of participants' own projects
- Focus on method choice, validation, and research design

# Introduction

Petro Tolochko

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- Research focus: text complexity, social networks, statistical modeling
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Noelle Lebernegg

- Post-Doc, Strategic Communication Science Group, Wageningen Univeristy
- Research focus: text as data, data mining, statistical modeling
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# Introduction: Participants

- Name?
- Disciplinary background?
- Experience with text analysis?
- Programming experience?
- Expectations / wishes for the workshop?

# Course Assessment

- In-class participation (20%)
- Final paper: application of one or several automated text analysis methods on a topic related to a topic of interest (e.g., related to your PhD thesis) (80%)
  - Content: short motivation, analysis (incl. commented code), description and interpretation of results (about 10 pages)
  - Format: Jupyter notebook (.ipynb) or R Markdown (.Rmd)
  - Deadline: 31. January, 2026
  - Project Talk: Day 4, afternoon

# Course Resources

Course repository:

- `https://github.com/noellelebe/text-as-data-2026/`
- Code, datasets, notebooks, and exercises will be organized there.

# Some Python Resources

- Van der Plas, J. (2026). Python Data Science Handbook: Essential Tools
  `https://github.com/jakevdp/PythonDataScienceHandbook`

# Some R Resources

- R for Data Science: `https://r4ds.had.co.nz/`
- Tidy Modeling with R: `https://www.tmwr.org/`
- Big Book of R. Database of R resources. `https://www.bigbookofr.com`

# R and Python Ressources for Text Analysis

- Atteveldt, W., Trilling, D., & Arcila, C. (2021). Computational Analysis of Communication. Wiley-Blackwell. `http://cssbook.net`
- Hugging Face. NLP Course. `https://huggingface.co/learn/nlp-course/chapter0/1`

# Books for the course

**Grimmer, Roberts & Stewart (2022)**
*Text as Data*
Research design and measurement with text.

**Jurafsky & Martin**
*Speech and Language Processing*
NLP methods and language models. Free third-edition draft.

**James et al. (2023)**
*An Introduction to Statistical Learning*
Python edition. Model fitting and evaluation. Free PDF.

# Further reading and practical references

**Manning, Raghavan & Schütze (2008)**
*Introduction to Information Retrieval*
Word counts, TF-IDF, and vector similarity. Free online.

**Cosma Rohilla Shalizi**
*Advanced Data Analysis*
*from an Elementary Point of View*
Statistical reasoning and model evaluation. Free draft, R examples.

# Today

- Text as data, text as measurement
- Document-term matrices and bag-of-words
- Preprocessing as an analytical choice
- TF-IDF and feature weighting
- Similarity, distance, and comparing texts
- Practical: how representation changes results

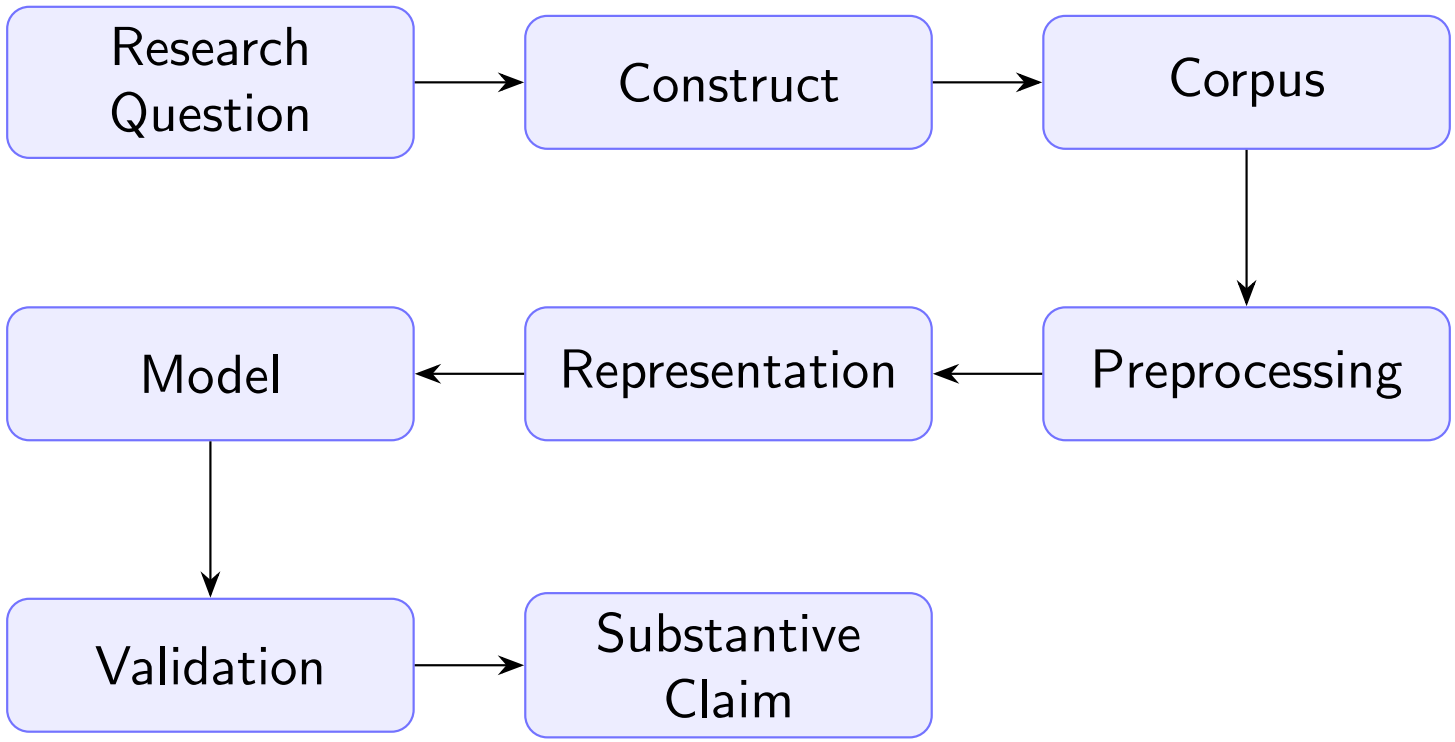
# Why Analyse Texts?

- Vast quantities of textual data are available
- Texts carry meaning
- Texts capture social, political, and cultural behaviour
- Texts can help us study phenomena that are otherwise difficult to observe

# What Do We Measure Through Text?

- We usually do not analyse text "for its own sake"
- We use text to measure latent or difficult-to-observe concepts:
  - attention sentiment framing ideology
  - complexity toxicity uncertainty polarization
- The representation determines what the model can measure

# The Text Analysis Pipeline



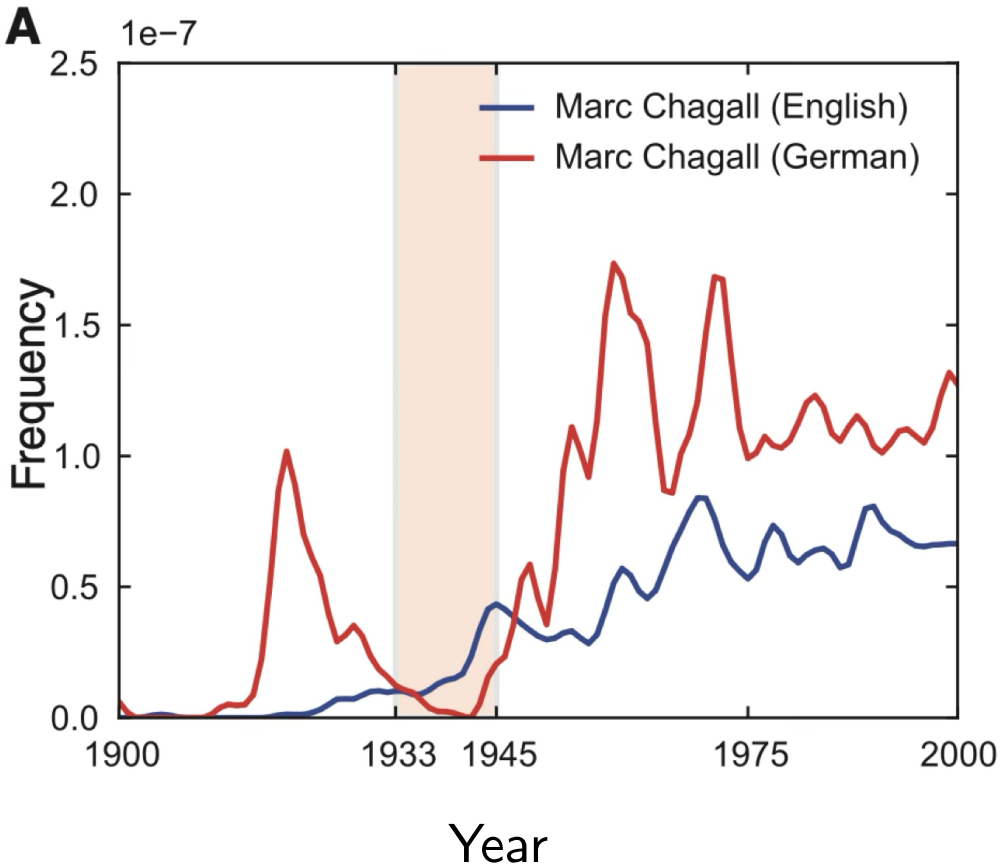
# Research questions and text

Texts record what people choose to say or write.

- Speeches can reveal political attention
- News articles can reveal how an issue is framed
- Open survey responses can reveal concerns that fixed questions miss

# Censorship in digitized books

Can changes in name frequency reveal censorship?




**Texts**
English and German books
**Measure**

Frequency of "Marc Chagall" by year **Pattern**

German mentions nearly disappear during Nazi rule

**Discuss:** What else could make a name disappear from books?

Source: Michel et al. (2011), *Science*, Figure 4A.

# Policy adoption and newspaper framing

Do policies adopted elsewhere predict how newspapers discuss smoking restrictions? **Texts:** over 52,000 newspaper paragraphs

49 US states, 1996–2013 **Result from the topic-model analysis**

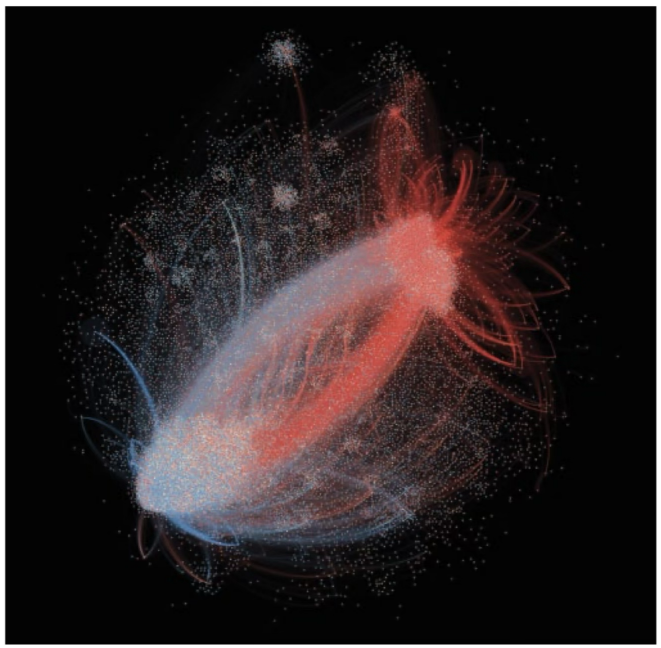
| **Framing of the policy** | **Earlier adoptions elsewhere** |
|---|---|
| Practical consequences | Predictive relationship |
| Normative justifications | No predictive relationship |

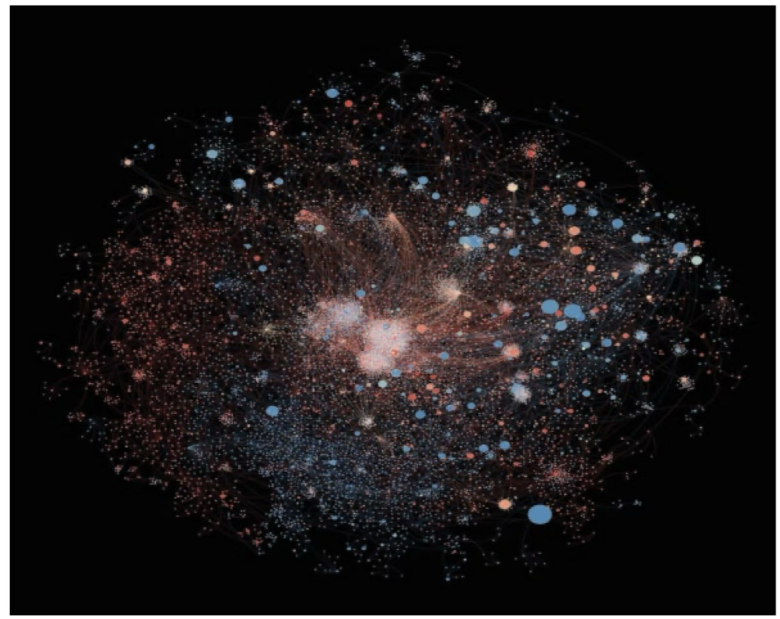
Source: Gilardi, Shipan & Wüest (2021), *AJPS*.

# Is online communication an echo chamber?

**Barberá et al. (2015):** does information travel across ideological boundaries? Nearly 150 million tweets on 12 political and nonpolitical issues;

estimated ideology for 3.8 million users.





Political issues showed more exchange among ideologically similar users; many nonpolitical events did not.

Source: Barberá et al. (2015), *Tweeting From Left to Right*, Psychological Science.

# First measurment?

How much attention do US presidents give to jobs in their State of the Union speeches? We will start by reading passages, then build and check a

simple word-count indicator.

# Jobs in presidential speeches

**Barack Obama, 2010**

> *"That is why jobs must be our number-one focus in 2010, and that's why I'm calling for a new jobs bill tonight."*

**Donald Trump, 2020**

> *"Jobs are booming, incomes are soaring, poverty is plummeting, crime is falling, confidence is surging, and our country is thriving and highly respected again."*

**Discuss:** Do these passages express the same thing about jobs?

# A concept and a first indicator

**Concept:** attention to employment in a speech **First indicator:**

occurrences of `job` or `jobs`

$$\text{mentions per 1,000 tokens} = \frac{\text{count of job + jobs}}{\text{number of tokens}} \times 1{,}000$$

- A longer speech has more opportunities to mention a word
- A rate makes speeches of different lengths easier to compare

**Discuss:** Which discussions of employment would this count miss?

# Checking the indicator against the text

**Another use of "jobs" in Obama's 2010 speech**

> *"That's why we've excluded lobbyists from policymaking jobs or seats on Federal boards and commissions."*

- Read passages that the indicator counts
- Look for relevant passages that it misses
- Decide what counts as attention to employment

**Discuss:** Would you include this passage? Why?

# Exercise: First text analysis project idea

- Form groups (of two to three people)
- Discuss initial text analysis project ideas

→ You can talk about project ideas that you already have or about areas in your discipline where large text quantities play a role

# From Text to Structure

# What is Text?

- Data
- Unstructured
- Highly multidimensional
- Difficult to work with if you are not human

# The Basic Problem

- Text is meaningful to humans
- But computers do not directly "understand" text
- We need to transform text into structured representations

# From Text to Structure

- We need to structure text before analysis
- We need representations that computers can process
- We also need representations that remain meaningful to us (and our research questions)
- Different research questions $\rightarrow$ different representations

# Unit of Analysis

| **Research question** | **Possible unit** |
|---|---|
| Does newspaper coverage become more negative? | article / paragraph / sentence |
| Do parties use simpler language? | speech / manifesto / paragraph |
| Are social-media posts toxic? | post / comment |
| What topics dominate climate discourse? | article / document / sentence |

- Unit choice determines what the model can see

# Document-Term Matrix

$$X = N \times K$$

- $N$ = number of documents
- $K$ = number of terms/features
- $X$ = structured representation of the corpus

# Document-Term Matrix

$$X = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$N = 3, \qquad K = 6$$

# Example Corpus

- Doc 1: "John loves ice-cream"
- Doc 2: "John loves oranges"
- Doc 3: "Mary hates ice-cream"

$$N = ?, \qquad K = ?, \qquad X = ?$$

# Document-Term Matrix with Vocabulary

| | John | loves | ice-cream | oranges | Mary | hates |
|---|---|---|---|---|---|---|
| $D_1$ | 1 | 1 | 1 | 0 | 0 | 0 |
| $D_2$ | 1 | 1 | 0 | 1 | 0 | 0 |
| $D_3$ | 0 | 0 | 1 | 0 | 1 | 1 |

$$N = 3, \qquad K = 6, \qquad X \in \mathbb{R}^{3 \times 6}$$

# Types & Tokens

- **Tokens:** all word occurrences
- **Types:** unique word forms

Example corpus:

- 9 tokens
- 6 types

# Bag-of-Words Representation

- Representation of text as a bag of words
- Word order is ignored
- Each document is represented as counts of words
- Simple, powerful, and still widely used

# What Bag-of-Words Captures and Loses

**Captures**

- word frequency
- topical emphasis
- repeated lexical choices
- broad document similarity

**Loses**

- word order
- syntax
- negation scope
- irony and sarcasm
- who did what to whom

# Bag-of-Words Is an Abstraction

- Bag-of-words is not "stupid"
- It deliberately discards some information
- It keeps information that is often useful:
  - lexical choice
  - frequency
  - relative emphasis
  - co-occurrence across documents
- Whether this abstraction is appropriate depends on the research question

# Sparse High-Dimensional Data

- Most corpora have many unique terms
- Most documents contain only a small fraction of all terms
- Result: document-term matrices are usually sparse
- This affects modeling, storage, computation, and interpretation

# Word counts as a probability

# Multinomial Model of Language

- Text $\approx$ draw from a Multinomial distribution

# Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- $n$ = number of trials
- $k$ = number of successes
- $p$ = success probability

# Multinomial Distribution

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$$

where: $n$ = text length, $k$ = vocabulary size, $p_i$ = probability of word $i$

# Binomial vs. Multinomial

- **Binomial:** one word (success/failure)

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

- **Multinomial:** multiple words/categories

$$P(x_1, \ldots, x_k) = \frac{n!}{x_1! \cdots x_k!} \prod_{i=1}^{k} p_i^{x_i}$$

# Example Texts

- Text 1 = banana banana banana banana chocolate
- Text 2 = chocolate chocolate chocolate banana fudge
- Text 3 = banana banana
- Text 4 = ice-cream ice-cream fudge ice-cream banana chocolate
- Text 5 = fudge fudge fudge chocolate
- Text 6 = ice-cream chocolate ice-cream fudge fudge fudge

# Example Texts

**John**
Text 1 = banana banana banana banana chocolate
Text 2 = chocolate chocolate chocolate banana fudge
Text 3 = banana banana

**Mary**
Text 4 = ice-cream ice-cream fudge ice-cream banana chocolate
Text 5 = fudge fudge fudge chocolate
Text 6 = ice-cream chocolate ice-cream fudge fudge fudge

# Document-Term Matrix

| | banana | chocolate | fudge | icecream |
|---|---|---|---|---|
| Doc 1 | 4 | 1 | 0 | 0 |
| Doc 2 | 1 | 3 | 1 | 0 |
| Doc 3 | 2 | 0 | 0 | 0 |
| Doc 4 | 1 | 1 | 1 | 3 |
| Doc 5 | 0 | 1 | 3 | 0 |
| Doc 6 | 0 | 1 | 3 | 2 |

# Document-Term Matrix

| | banana | chocolate | fudge | icecream |
|---|---|---|---|---|
| John Rates | 7 | 4 | 1 | 0 |
| Mary Rates | 1 | 3 | 7 | 5 |

# Language Models

## John Language Model

| | | | | |
|---|---|---|---|---|
| John Rates | 7 | 4 | 1 | 0 |

## Mary Language Model

| | | | | |
|---|---|---|---|---|
| Mary Rates | 1 | 3 | 7 | 5 |

# Language Models

## John Language Model

| | | | | |
|---|---|---|---|---|
| John Rates | $\frac{7}{12}$ | $\frac{4}{12}$ | $\frac{1}{12}$ | $\frac{0}{12}$ |

## Mary Language Model

| | | | | |
|---|---|---|---|---|
| Mary Rates | $\frac{1}{16}$ | $\frac{3}{16}$ | $\frac{7}{16}$ | $\frac{5}{16}$ |

# New Texts

- `new_text_1` = ice-cream fudge fudge
- `new_text_2` = chocolate chocolate banana banana

Question: Probability they were generated by John or Mary?

# New Texts Document-Term Matrix

| | banana | chocolate | fudge | icecream |
|---|---|---|---|---|
| New Text 1 Rates | 0 | 0 | 2 | 1 |
| New Text 2 Rates | 2 | 2 | 0 | 0 |

# Probability Spoken by John

- `new_text_1` = "ice-cream fudge fudge"

$$\Pr(b = 0, ch = 0, f = 2, i = 1) = \frac{3!}{0!0!2!1!} \times (0.583)^0(0.333)^0(0.08)^2(0)^1 = 0$$

# Probability Spoken by John

- `new_text_2` = "chocolate chocolate banana banana"

$$\begin{aligned}\Pr(b=2, ch=2, f=0, i=0) &= \frac{4!}{2!2!0!0!}(0.583)^2(0.333)^2(0.08)^0(0)^0 \\ &= 6(0.583)^2(0.333)^2 \\ &\approx 0.23\end{aligned}$$

# Probability spoken by Mary

- `new_text_1` (Mary): $\Pr = 0.18$
- `new_text_2` (Mary): $\Pr = 0.0008$

# Documents as vectors

# Vector Space Model

- Represent texts as vectors in multidimensional space
- Each dimension corresponds to a feature
- In a document-term matrix, features are usually words or phrases
- Once texts are vectors, we can compare them geometrically

# Multidimensional Spaces

Multidimensional?

- 2D: position on a map
  $X$ = longitude, $Y$ = latitude
- 3D: position in the real world
  $X$ = longitude, $Y$ = latitude, $Z$ = height
- 4D:
  $X$ = longitude, $Y$ = latitude, $Z$ = height, $T$ = time

# Multidimensional Spaces

- The number of dimensions is the number of features needed to describe an object
- In geography: coordinates describe location
- In text analysis: words/features describe the position of a document in linguistic space

# Two-Dimensional Space

Vocabulary = {"banana", "chocolate"}

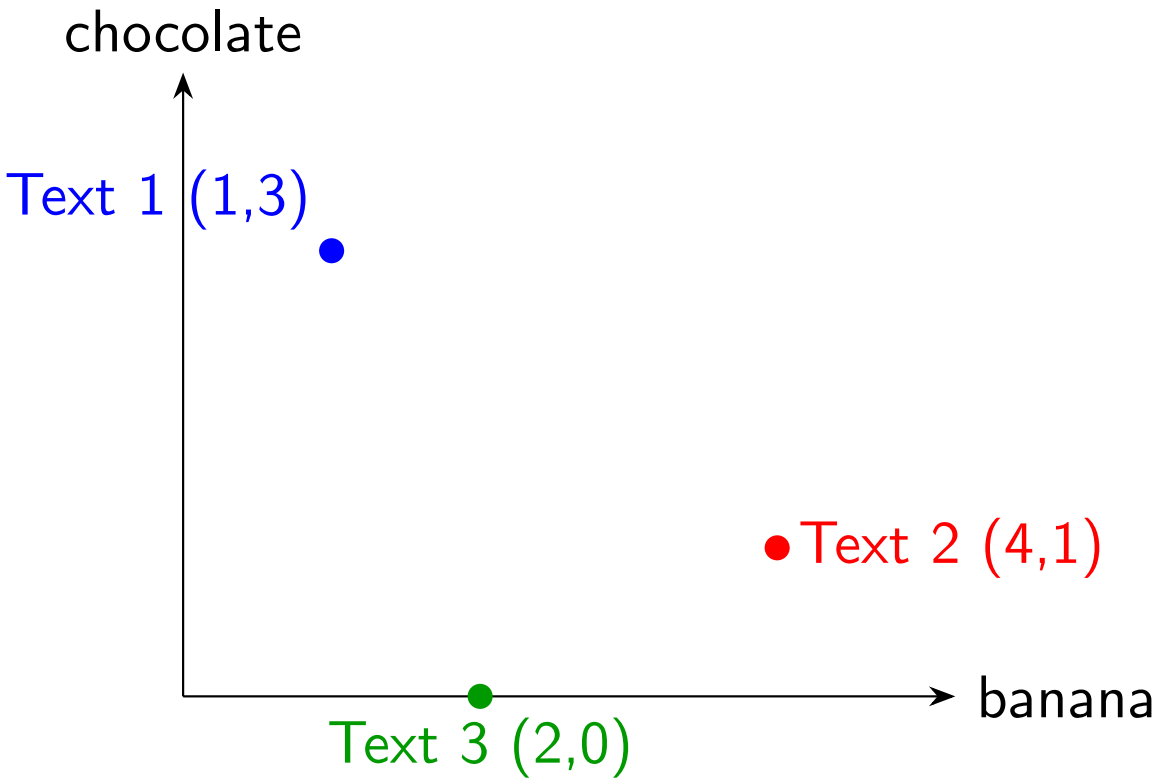
- Text 1 = "chocolate chocolate chocolate banana"
- Text 2 = "banana banana banana banana chocolate"
- Text 3 = "banana banana"

# Two-Dimensional Space

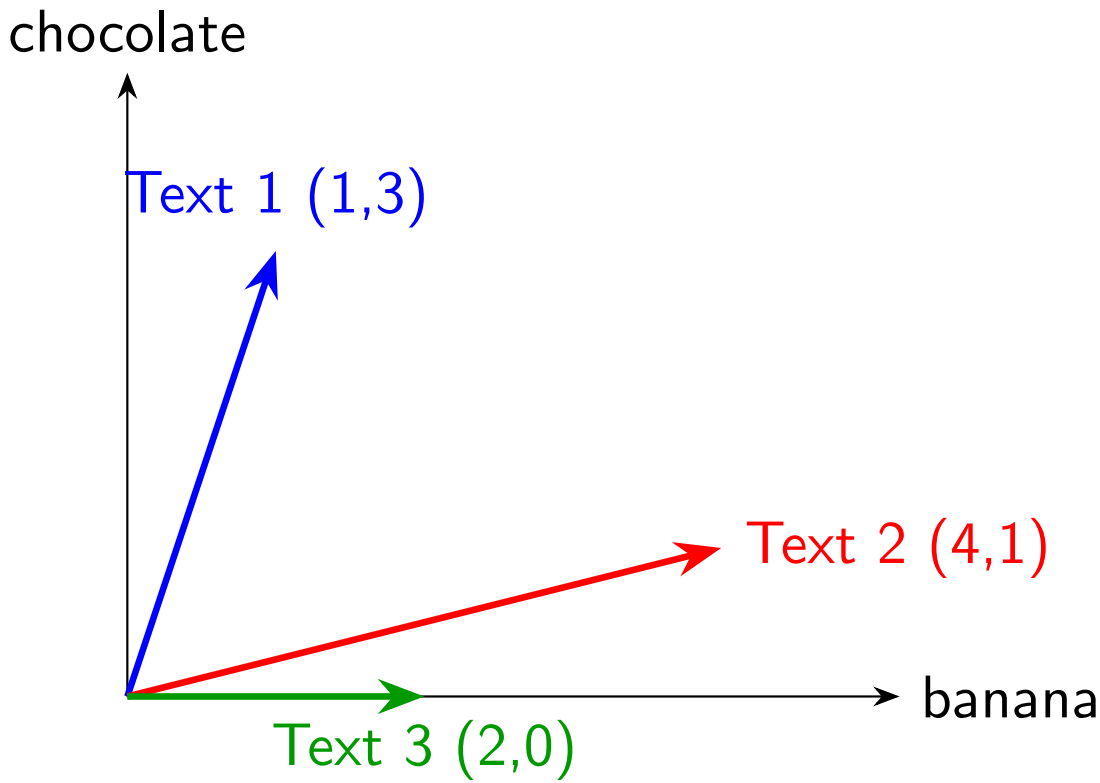
Vocabulary $= \{$"banana", "chocolate"$\}$

- Text 1 = "chocolate chocolate chocolate banana" $\Rightarrow (1, 3)$
- Text 2 = "banana banana banana banana chocolate" $\Rightarrow (4, 1)$
- Text 3 = "banana banana" $\Rightarrow (2, 0)$

# Two-Dimensional Vector Space



# Two-Dimensional Vector Space



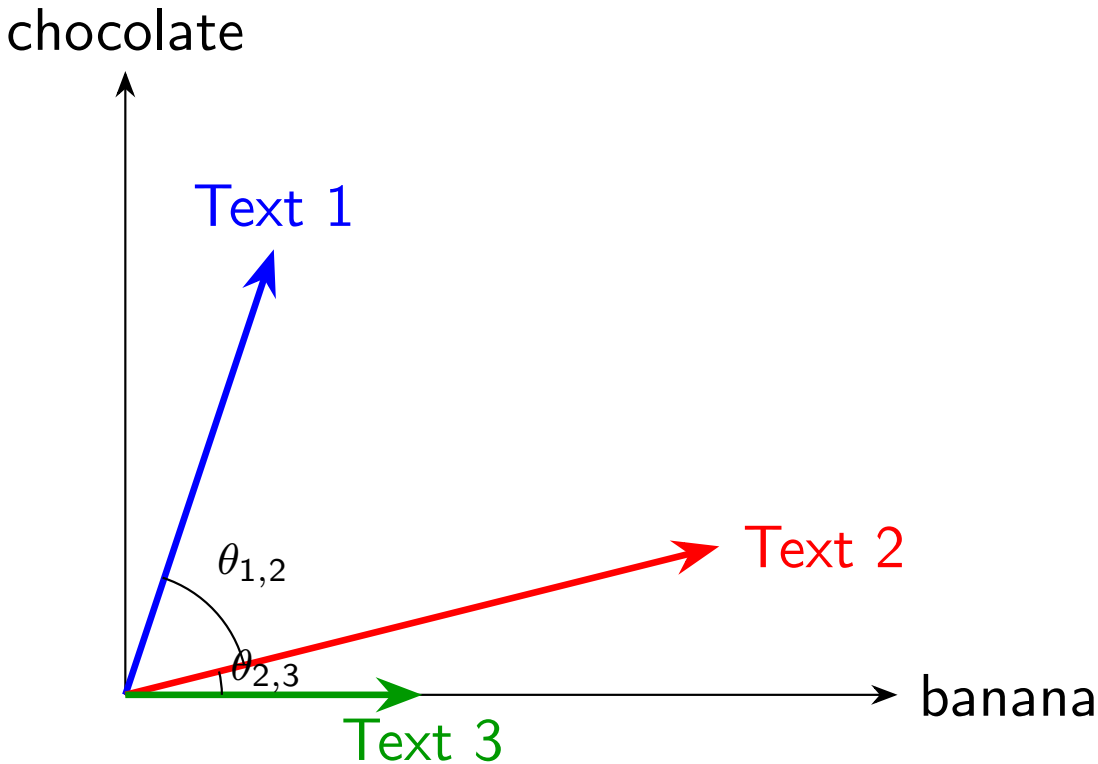
# Cosine Similarity

$$\cos(\theta) = \frac{x \cdot y}{\|x\|\|y\|}$$

- Measures the angle between two vectors
- Commonly used for document similarity
- Less sensitive to document length than raw dot products

$$\cos(\theta_{1,2}) = \frac{1 \cdot 4 + 3 \cdot 1}{\sqrt{1^2 + 3^2}\sqrt{4^2 + 1^2}} = \frac{7}{\sqrt{10}\sqrt{17}} \approx 0.538$$

# Two-Dimensional Vector Space with Angles



# Vector Distances

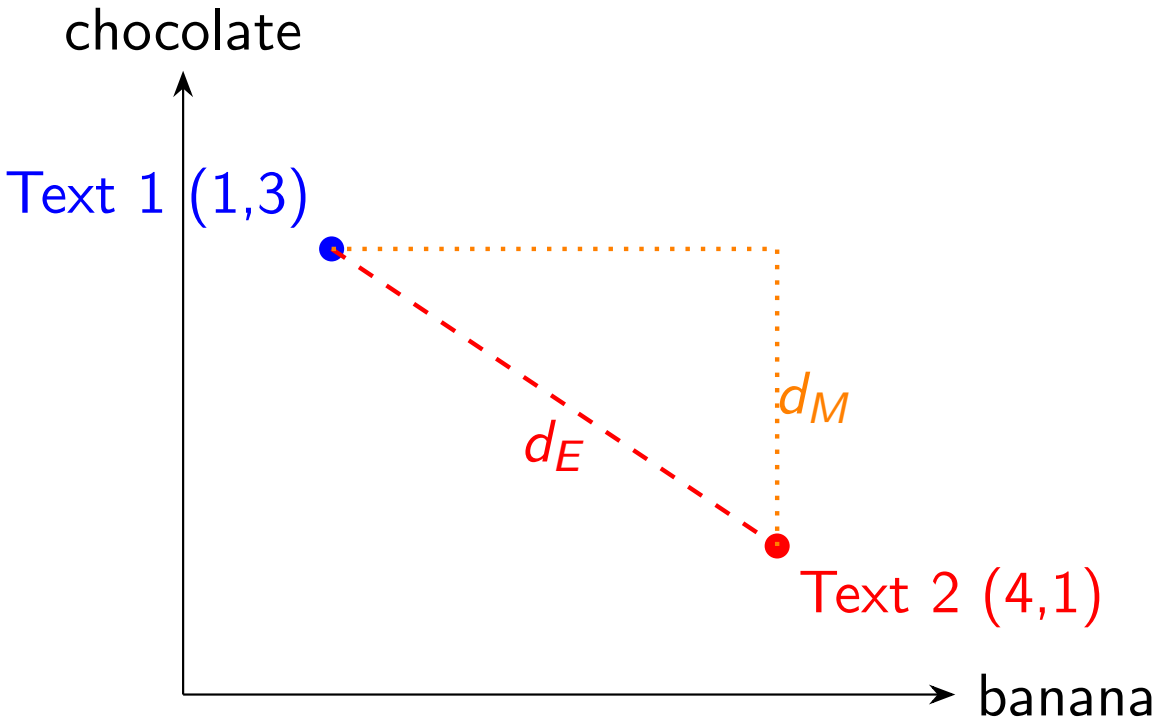
$$d_E(x, y) = \sqrt{\sum_i (x_i - y_i)^2} \qquad d_M(x, y) = \sum_i |x_i - y_i|$$

For Text 1 $(1, 3)$ and Text 2 $(4, 1)$:

$$d_E(T1, T2) = \sqrt{(1-4)^2 + (3-1)^2} = \sqrt{13} \approx 3.61$$

$$d_M(T1, T2) = |1-4| + |3-1| = 5$$

# Euclidean vs. Manhattan Distance



# From Two Dimensions to Thousands

- Real vocabularies are large
- A document-term matrix can easily have tens of thousands of columns
- Same logic as the two-dimensional example
- Same math, but no longer visually intuitive

# Feature engineering & preprocessing

# Feature Engineering

- A **feature** is a property of a document recorded in the data matrix.

$$X_{ij} = \text{value of feature } j \text{ for document } i$$

- **Which columns?** Single words, phrases, or other properties of the text.
- **What values?** Counts, 0/1 presence, or weights such as TF–IDF.

These choices define what the matrix can capture.

# Which Columns Are in the Matrix?

**Two documents**

D1: "climate change matters" D2: "climate policy matters"

| **Choice** | **Possible columns** |
|---|---|
| Unigrams | climate, change, policy, matters |
| Add bigrams | climate_change, change_matters, climate_policy, policy_matters |
| Require `min_df=2` | climate, matters (each appears in both documents) |

Adding phrases creates columns.

# What Goes in Each Cell?

**Two documents**

D1: "climate climate policy" D2: "climate policy"

| | **Counts** | | **Binary presence** | |
|---|---|---|---|---|
| | climate | policy | climate | policy |
| D1 | 2 | 1 | 1 | 1 |
| D2 | 1 | 1 | 1 | 1 |

- Binary values discard repetition within a document.
- TF–IDF rescales counts using how widely terms occur in the corpus.

# Text Preprocessing

- Texts are highly dimensional
- Preprocessing reduces dimensionality and standardizes features
- But preprocessing is not neutral
- It decides which distinctions are signal and which are noise

# Common Preprocessing Choices

- Punctuation
- Numbers
- Lowercasing
- Stemming / lemmatization
- Stop-word removal
- N-grams
- Removal of words by frequency

Based on common preprocessing decisions discussed in text-as-data workflows, including Denny & Spirling (2018).

# Punctuation / Numbers / Lowercasing

- Often we remove punctuation and/or numbers
- Often we lowercase all text
- This reduces dimensionality
- But it can remove meaningful distinctions

Is Trump the same as trump?
Is `Article 5` meaningful?
Does `!!!` matter in social media text?
Is `COVID-19` just noise?

# Lowercasing

**Can help**

- reduces vocabulary size
- combines accidental variants
- useful for many BoW tasks

**Can hurt**

- names and entities
- acronyms
- German nouns
- stylistic emphasis

# Stemming / Lemmatization

- A stem is the part of the word responsible for lexical meaning
- A stem is the invariable part of the word under inflection
- "wait" is a stem of:
  - waiting
  - waits
  - waited
- A **lemma** is the base or dictionary form
  - "went", "gone" $\Rightarrow$ "go"

# Stemming / Lemmatization: Why It Matters

- Useful for dimension reduction
- Often helps when measuring broad lexical content
- But it can erase meaningful stylistic or grammatical distinctions

## Question

When would you want to keep inflected forms separate?

# Stop Words

- Words filtered out before analysis begins
- Usually function words:
  - the
  - is
  - that
  - and
- Domain-specific words are also often excluded
  - e.g., "climate" in a corpus about climate change

# Stop Words Are Not Always Useless

- Function words may be crucial for authorship attribution
- Pronouns may matter for populism, identity, or blame attribution
- Negation words matter: not, never, no
- Modal verbs may matter: must, should, can

# N-grams

- So far, we have mostly used unigrams: individual words
- Texts can also be represented as sequences of *n* words
- Example: "I love ice-cream and bananas"
  - unigrams: "I", "love", "ice-cream", "and", "bananas"
  - bigrams: "I love", "love ice-cream", "ice-cream and", "and bananas"

# Why N-grams?

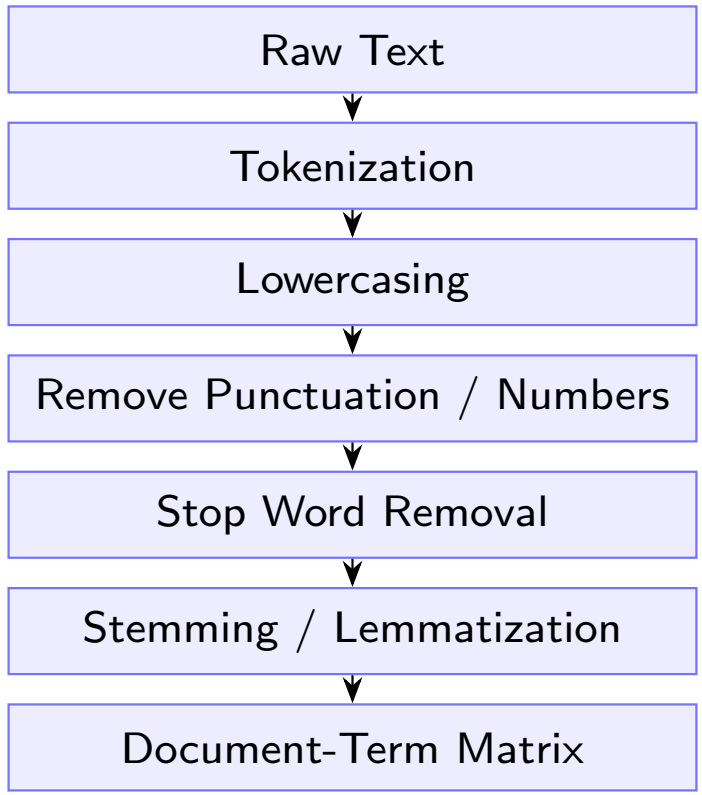
| **Unigrams** | **N-gram meaning** |
|---|---|
| climate, change | climate change |
| fake, news | fake news |
| not, good | not good |
| European, Union | European Union |
| social, media | social media |

- N-grams partly recover information lost by bag-of-words

# Removal of Terms by Frequency

- Further dimensionality reduction can be achieved by removing:
  - very frequent terms
  - very infrequent terms
- Very frequent terms may not discriminate between documents
- Very infrequent terms may have low statistical power
- But rare terms can still be substantively important

# Preprocessing Order Matters



# Preprocessing Changes the Estimand

- Suppose we remove punctuation, numbers, stop words, and word order
- We are no longer measuring "language use" in general
- We are measuring lexical content after discarding:
  - style  syntax  discourse markers
  - many forms of emphasis  some forms of negation
- This may be exactly what we want, but it should be a choice, not a default

# Raw counts and TF-IDF

# Beyond Raw Counts

- A document-term matrix usually contains raw counts
- But raw counts are not always ideal
- Very frequent words dominate
- Longer documents have larger counts
- Some words are common everywhere and not very distinctive

# TF-IDF

- We can do more than just count words
- We can transform counts using weights
- TF-IDF stands for:
  - term frequency
  - inverse document frequency
- Intuition: important terms are frequent in a document but rare across the corpus

# TF-IDF: Term Frequency – Inverse Document Frequency

- **TF:** how often a term appears in a document
- **IDF:** downweight terms that appear in many documents

$$\text{TF-IDF}(t, d) = \underbrace{\frac{\text{count}(t, d)}{\sum_{t'} \text{count}(t', d)}}_{\text{TF}} \times \underbrace{\log \frac{N}{\text{df}(t)}}_{\text{IDF}}$$

- $N$ = total number of documents
- $\text{df}(t)$ = number of documents containing term $t$

# TF-IDF: What the hell?

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# TF-IDF: What the hell?

- It looks arbitrary because it partly is
- Introduced by Karen Spärck Jones in 1972
- Not derived from a deep theory of meaning
- It is a useful heuristic: frequent in this document, rare elsewhere
- Often works surprisingly well as a baseline

# What TF-IDF Is Useful For

- Search and information retrieval
- Finding characteristic words in documents
- Document similarity
- Clustering
- Baseline features for supervised classification

# What TF-IDF Is Not

- It is not a direct measure of how much a concept appears
- It downweights common words even if they are theoretically important
- It changes the interpretation of word counts
- A high TF-IDF word is distinctive, not necessarily substantively important

# Comparing documents and groups

# Comparing Texts

Once we represent texts as vectors, we can compare them.

- Which documents are most similar?
- Which documents are outliers?
- Which documents cluster together?
- Which documents are closest to a query?

# Comparing Groups

Sometimes we do not only compare documents.

- Which words are distinctive of one group?
- Which words distinguish one corpus from another?
- Which words are associated with one party, outlet, period, or actor?

Examples:

- tabloids vs. broadsheets
- left parties vs. right parties
- before vs. after an event
- pro-government vs. opposition actors

# Odds

- **Probability:** $p = \frac{\#\text{ successes}}{\#\text{ trials}}$
- **Odds:** ratio of probability of success to probability of failure

$$\text{Odds} = \frac{p}{1-p}$$

If $p = 0.8$, then:

$$\text{Odds} = \frac{0.8}{0.2} = 4$$

4-to-1 in favor

# Log Odds

- Log odds are the natural logarithm of odds
- They map probabilities $(0, 1)$ to the full real line $(-\infty, +\infty)$

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right)$$

If $p = 0.8$, then:

$$\log\left(\frac{0.8}{0.2}\right) = \log(4) \approx 1.39$$

# Comparing Two Groups

- Suppose we have two groups, for example John and Mary
- Each group has its own probability of using a word
- We compare their log odds

$$\text{Log Odds Ratio} = \log\left(\frac{p_1}{1-p_1}\right) - \log\left(\frac{p_2}{1-p_2}\right)$$

- Positive value: word is more associated with group 1
- Negative value: word is more associated with group 2

# Worked Example: "banana"

- John says "banana" in 7 out of 12 words: $p_1 = 7/12 \approx 0.58$
- Mary says "banana" in 1 out of 16 words: $p_2 = 1/16 = 0.0625$

$$\begin{aligned}\text{Log Odds Ratio} &= \log\left(\frac{0.58}{0.42}\right) - \log\left(\frac{0.0625}{0.9375}\right) \\ &= \log(1.38) - \log(0.0667) \\ &\approx 0.32 - (-2.71) \approx 3.03\end{aligned}$$

Positive log-odds ratio $\Rightarrow$ "banana" is more associated with John.

# Why Use Log Odds?

- Probabilities are bounded between 0 and 1
- Odds ratios can be very skewed
- Log odds:
  - are symmetric around 0
  - have direction and magnitude
  - are related to logistic regression
  - are useful for comparing word usage between groups
- In text analysis, log-odds approaches are often used to find distinctive words

# Choosing a representation

# Representation Choice Matrix

| **Goal** | **Useful representation** | **Risk** |
|---|---|---|
| Measure word prevalence | raw counts / proportions | common words dominate |
| Find distinctive terms | TF-IDF / log odds | distinctiveness $\neq$ importance |
| Compare documents | TF-IDF + cosine similarity | length and preprocessing matter |
| Capture phrases | n-grams | feature explosion |
| Reduce dimensionality | trimming / stemming | loss of meaning |
| Measure style | keep function words / punctuation | noisy features |

# Representation Choices Are Research Choices

- If you remove stop words, you probably cannot study style well
- If you remove numbers, you may lose dates, prices, laws, or percentages
- If you use unigrams, you may miss fixed phrases
- If you use TF-IDF, you are measuring distinctiveness, not raw prevalence
- If you aggregate to documents, you may lose sentence-level meaning

# No Universal "Correct" Preprocessing

## Bad question

What is the correct preprocessing pipeline?

## Better question

What preprocessing pipeline is appropriate for this research question, corpus, and validation strategy?

# Beyond Bag-of-Words

- Bag-of-words represents documents by explicit word counts.
- This is simple, transparent, and often useful.
- But it treats words as independent features.
- It does not know that:
  - "car" and "automobile" are similar
  - "good" and "not good" are different
  - the same word can mean different things in different contexts
- Later in the course, we will move from count-based representations to learned representations.

# Distributional Hypothesis

- **Key idea:** words that occur in similar contexts tend to have similar meanings.
- "You shall know a word by the company it keeps."
- This motivates representing words by their contexts.
- Instead of asking only how often a word appears, we ask:
  - Which words appear near it?
  - Which documents does it appear in?
  - Which contexts is it associated with?

# Preview: Word Embeddings

- Word embeddings represent words as dense vectors.
- Similar words should be close together in vector space.
- Unlike bag-of-words, the dimensions are learned and not directly interpretable.
- Embeddings are useful for:
  - semantic similarity
  - measuring concepts with word lists
  - document representations
  - transformers and LLMs
- We return to this in detail on Day 4.

# Practical session

# Practical Session: Guided Walkthrough

- Work through notebook sections 1–8 together.
- Start with a small toy corpus, then use State of the Union speeches.
- Build word counts and TF–IDF representations.
- Inspect distinctive words and document similarity.

**Then:** Section 9 is your own short comparison.

# Your Turn: Change One Representation

In notebook section 9, change only `task_choice`. Use the same speeches.

- `min_df`: keep terms appearing in at least 5 speeches instead of 2.
- `binary`: count whether a term appears in a speech (0/1), not repetitions.
- `bigrams`: keep single words and add adjacent word pairs.

Choose **one** option and rerun the student-task cell.

# Compare the Two Versions

The cell prints a baseline and your changed representation.

- How many features does each version have?
- Which terms were added or removed?
- Did the top terms change? What do the totals mean in each version?
- Which recent speech pairs changed most in cosine similarity?

**Remember:** A small change in top terms can still hide a large change in the vocabulary or similarities.

# Discussion: What Did Your Choice Do?

Write three sentences:

1. What changed between baseline and your version?
2. Why did this representation produce that change?
3. Which version better fits a research question you care about?

**Optional extension:** Compare speeches from 1950–79 with those from 1980 onward. This changes the groups being compared, rather than the representation.

# Key Takeaways

- Text analysis begins with representation
- A document-term matrix is simple but powerful
- Bag-of-words is not stupid: it is a deliberate abstraction
- Preprocessing is not cleaning: it is measurement design
- TF-IDF and log odds help identify distinctive language
- Different representations support different claims

# The Main Lesson

There is no raw text analysis.

There is always a representation.

A representation decides what counts as signal, what counts as noise, and what kind of claim the analysis can support.

# Tomorrow

- Dictionaries and regular expressions
- Human annotation and codebooks
- Supervised machine learning
- Evaluating classification models